

ScreenSense Guardian – Complete Project Summary

The On-Device AI Copilot for Every Screen

1. The Core Problem

Gen Z grew up with intuitive tech. Everyone else is trying to keep up.

People who are 40+ today – and especially elderly users – grew up with technology, but not with *today’s* technology. Interfaces, security models, and AI tools changed under their feet. Gen Z finds certain things effortless because they were born into them. 40+ users have deep life and work experience, but often feel left behind.

The gap is not intelligence. It’s exposure and interface design.

Gen Z does effortlessly	40+ and elderly often struggle with
Cloud sync, QR codes, 2FA, password managers	Too many passwords, account lockouts, 2FA resets
Spotting scams, deepfakes, phishing	Falling for fake bank SMS, AI voice calls, tech support pop-ups
Prompting AI, using voice assistants	Not knowing how to ask or where to start
Organizing photos, files, documents	Messy archives across old phones, USB drives, cloud
Navigating government and health portals	Confusing forms: insurance, taxes, medical, benefits
Multitasking across apps and devices	“How do I attach a file?” in Gmail, Teams, WhatsApp
Using GitHub, VS Code, terminal	Intimidated by developer tools and command line
Using Word, PowerPoint, Excel	Forgetting where a button is, watching long YouTube tutorials
Hearing fast speech and accents	Missing key points in calls, feeling left out

These are not intelligence problems. They are **exposure and interface problems**. ScreenSense Guardian is the patient bridge.

2. The Solution

ScreenSense Guardian is an on-device AI layer that watches your screen (with permission), understands what you’re looking at, and helps you in plain language – instantly, privately, and offline.

It runs entirely on the Snapdragon NPU inside an HP PC. Nothing is sent to the cloud. No data leaves your laptop.

It has **two modes** that work together:

Mode	Purpose
Guide Mode	Helps with any task: “How do I attach a file?” “Where did my download go?” “How do I preview a markdown file?” “How do I add a picture in PowerPoint?”
Guardian Mode	Detects scams, phishing, fake pop-ups, suspicious websites, and warns the user in real time.

Same app. Same models. Same privacy. Same low latency.

3. What ScreenSense Guardian Does

A. Government & Health Portals

The AI can:

- Explain jargon: “Adjusted gross income means...”
- Say which documents are needed: “You’ll need your W-2 and last year’s return.”
- Auto-fill from a local profile with permission.
- Warn: “This site is not the official government domain.”
- Check before submit: “You missed the signature field.”
- Remind: “This deadline is March 15.”

Gen Z breezes through these. Many 40+ users give up or pay someone. This closes that gap.

B. Any Website

Because it’s screen-aware, not site-specific, it can:

- Summarize long articles or terms of service.

- Explain cookie settings and privacy choices.
- Detect dark patterns: “This button signs you up for a subscription.”
- Warn about phishing: “This login page looks fake.”
- Help with 2FA: “Open your authenticator app and enter the 6-digit code.”
- Translate confusing language into plain English.

Works on news sites, banking, insurance, travel, shopping, medical portals — anything.

C. Scam & Fraud Detection (Guardian Mode)

Real scenarios it can catch:

Scenario	How AI responds
Fake virus pop-up: “Your PC is infected! Call Microsoft now!”	“This is a tech support scam. Do not call. Close this window.”
Phishing email: “Your bank account is locked. Click here.”	Reads email, checks link, explains red flags.
Fake bank SMS: “Share your OTP to verify.”	“Banks never ask for OTP. Do not reply.”
Grandparent scam call: “Grandma, I’m in trouble, send money.”	Transcribes call, detects script, warns user.
Fake Medicare website	Looks at page layout and logos. “This is not the official site.”

D. GitHub & Developer Workflows

This is where the “show, don’t do” philosophy shines.

The AI does NOT run commands. It shows the user exactly what to do, where to press, and what to copy.

User asks	AI response
"How do I preview this .md file in VS Code?"	Highlights the Preview icon, shows <code>Ctrl+Shift+V</code> , and gives the Command Palette path.
"How do I set my git username and email?"	Shows terminal command box with copy button: <pre>git config --global user.name "Your Name" git config --global user.email "you@example.com"</pre>
"What should my commit message be?"	Reads the diff and suggests: Add user login validation and error handling with 2–3 alternative options.
"Why did my GitHub Actions workflow fail?"	Reads the error, explains it in plain English.
"How do I fork this repo?"	Step-by-step guidance with highlighted buttons.
"How do I resolve this merge conflict?"	Walks through it step by step.
"Is this repo safe to clone?"	Checks reputation and red flags.

Every command comes with a one-line explanation:

```
git config --global user.name "Your Name"
```

Sets your name for all future Git commits. Safe. Does not change any files.

That one line removes the fear.

E. Learn Any App — Word, PowerPoint, Excel, Windows, and More

This is the most relatable use case of all. It turns ScreenSense into a **patient, always-available tutor** that sits right on your screen.

No YouTube. No searching. No pausing and rewinding. No ads. No wrong-version tutorials. Just: **"Show me where to click, right here, right now."**

MS Word

User asks	AI does
“How do I add page numbers?”	Highlights Insert → Page Number , shows the dropdown, says “Pick where you want it — top or bottom.”
“How do I make this text bold?”	Highlights the B button, says “Or press <code>Ctrl+B</code> .”
“How do I insert a table?”	Highlights Insert → Table , says “Drag to pick rows and columns.”
“How do I change margins?”	Highlights Layout → Margins , shows the preset options.
“How do I save as PDF?”	Highlights File → Save As → PDF , gives keyboard shortcut.
“Why is my formatting messed up?”	Reads the screen, spots the issue, explains: “You have extra paragraph breaks. Want me to show where?”

MS PowerPoint

User asks	AI does
“How do I add a picture?”	Highlights Insert → Pictures → This Device , says “Then pick your photo.”
“How do I make a new slide?”	Highlights New Slide , or says “Press <code>Ctrl+M</code> .”
“How do I add animations?”	Highlights Animations tab , shows the options, explains “Fade is safest for work.”
“How do I make text bigger?”	Highlights font size box, says “Or select text and press <code>Ctrl+Shift+></code> .”
“How do I present this?”	Says “Press <code>F5</code> to start from the beginning.”
“How do I change the theme?”	Highlights Design tab , shows theme thumbnails.

MS Excel

User asks	AI does
“How do I add up a column?”	Shows <code>=SUM(A1:A10)</code> in a copy box, highlights the formula bar.
“How do I make a chart?”	Highlights Insert → Chart , says “Select your data first.”
“How do I freeze the top row?”	Highlights View → Freeze Panes , says “Pick ‘Freeze Top Row’.”
“How do I sort this?”	Highlights Data → Sort , explains A-Z vs Z-A.
“What does this error mean?”	Reads <code>#REF!</code> or <code>#VALUE!</code> , explains in plain English.

Windows Itself

User asks	AI does
“How do I take a screenshot?”	Says “Press <code>Win+Shift+S</code> . Then drag to select.”
“Where did my download go?”	Opens File Explorer, highlights the Downloads folder.
“How do I connect to Wi-Fi?”	Highlights the Wi-Fi icon in the taskbar.
“How do I make text bigger?”	Highlights Settings → Display → Scale .
“How do I unzip this file?”	Highlights the file, says “Right-click → Extract All.”

Any App

User asks	AI does
“How do I attach a file in Gmail?”	Highlights the paperclip icon.
“How do I share my screen in Teams?”	Highlights the Share button, explains the options.
“How do I print this?”	Says “Press <code>Ctrl+P</code> . Pick your printer. Click Print.”
“How do I copy this?”	Says “Select it, then press <code>Ctrl+C</code> . Paste with <code>Ctrl+V</code> .”

F. Why This Beats YouTube

YouTube / Google	ScreenSense Guardian
You have to search and guess the right keywords	You just ask in plain language
Video shows a different version of Word/PowerPoint	AI sees <i>your</i> exact screen and version
You pause, rewind, lose your place	It waits for you. One step at a time.
Ads, intros, long videos	Instant answer
You have to switch tabs	It overlays on the same screen
No privacy — Google sees your search	Fully on-device. Nobody knows what you’re learning.
You feel dumb for searching “how to add picture in ppt”	It feels like a patient friend sitting next to you

The last one matters most. **Dignity.** 40+ users often avoid asking for help because they feel embarrassed. A private on-device tutor removes that shame.

4. The “Show, Don’t Do” Philosophy

This is a deliberate product decision, not a limitation.

Principle	Why it matters
AI never runs commands	User stays in control. No risk of destructive actions.
User copies and pastes	User learns while doing. Feels capable, not dependent.
No OS automation needed	Buildable without admin permissions or deep system hooks.
Easy to demo	Highlight button → show command → user pastes. Clean 2-minute video.
Privacy-safe	NPU reads screen locally. Nothing sent to cloud.
One step at a time	Don't dump 10 instructions. Show one step, wait, then show the next. Feels like a patient tutor, not a manual.

5. Design Rules for Elderly & 40+ Users

If you scare them, they won't use it.

1. **Calm, not alarmist** — “This might be a scam. Let's check together.”
2. **Always explain why** — “It uses urgent words and asks for money. Real banks don't do that.”
3. **Suggest a safe action** — “Call your bank using the number on your card.”
4. **Never auto-block without permission** — Give a big “Ignore” button.
5. **Large text, voice, simple words** — No jargon.
6. **One step at a time** — Don't overwhelm with 10 instructions.

6. Technical Architecture

User asks by voice or text

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Whisper-Small → transcribes the question

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Qwen2.5-VL-3B → reads the current screen + app

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Phi-3.5-mini → reasons about the step and writes the explanation

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Overlay UI → highlights the button + shows the command box

↓

Piper TTS → speaks the instruction aloud

↓

All running on Snapdragon Hexagon NPU (QNN EP)

Everything stays on the laptop. No cloud. No data leaks.

Optional add-on: A small local knowledge base of common how-to steps (Word, Excel, PowerPoint, Windows, Gmail, Teams). The LLM uses it as context so answers are accurate and version-aware. This is just a vector store — **bge-small-en-v1.5** handles it.

Production runtime:

```
session = ort.InferenceSession(  
    "snapdragon_optimized_model.onnx",  
    providers=['QNNExecutionProvider']  
)
```

7. AI Hub Models Used (All Free & Pre-Optimized)

Job	Model
Read text on screen	TrOCR / PaddleOCR
Understand screenshots & UI	Qwen2.5-VL-3B
Reason about scams & explain	Phi-3.5-mini or Llama-3.2-3B
Transcribe calls / voicemails	Whisper-Small
Compare against known scam patterns	bge-small-en-v1.5
Speak warnings aloud	Piper TTS
Code-specific help (optional)	CodeLlama or Phi-3.5-mini

All are open-weight, free, and available on Qualcomm AI Hub with pre-compiled QNN binaries for Snapdragon X Elite / X Plus.

8. Why Snapdragon NPU Matters

Requirement	Why cloud AI fails	Why Snapdragon NPU wins
Privacy	Screens, emails, SMS, call audio sent to server	Everything stays on device
Latency	500ms–2s roundtrip	<100ms local inference
Offline	Needs internet	Works on a plane, in a bunker
Battery	CPU/GPU drain, fans spin	Hexagon NPU handles audio + vision + language under 5W
Cost	API bills add up	Zero cloud API costs

9. What ScreenSense Guardian Is NOT

- Not a full IDE or code editor
- Not a replacement for Stack Overflow

- Not an auto-run tool
- Not a cloud service
- Not trying to handle every possible app

Focused scope: Screen reading + highlight + command suggestion + copy button + plain-English explanation.

That’s the product. That’s enough to win.

10. How It Fits the Qualcomm Challenge Rubric

Evaluation Pillar	How ScreenSense Guardian Scores
Technical Implementation	Uses QNN EP, INT8/W4A16 quantization, on-device vision + language + audio. Mentions specific runtimes: ONNX Runtime with QNN EP (QnnHtp.dll), Qualcomm AI Engine Direct SDK (QAIRT), Genie runtime.
Use Case & Innovation	High social impact. Solves real 40+ and elderly pain: scams, government forms, developer tools, learning Word/PowerPoint/Excel. Emotional hook.
Deployment & Accessibility	Fully offline, low latency, works on battery, private by design. Architecture diagram shows pipeline clearly.
Presentation & Documentation	Clear README, demo video, code quality, reproducibility. Emotional story. Easy to demo.

11. Submission Checklist

Section	What to include
Problem	Real 40+ and elderly pain: passwords, scams, forms, photos, calls, developer tools, learning Word/PowerPoint/Excel
Solution	ScreenSense Guardian running on Snapdragon HP PC with Guide Mode + Guardian Mode
Architecture	Input → Preprocessing → QNN/NPU → UI
AI Hub Models	Name each model (Qwen2.5-VL, Phi-3.5, Whisper, etc.) and link to AI Hub pages
Snapdragon Advantage	Privacy, low latency, offline, battery efficiency
Demo	Screenshots or short video: fake pop-up warning, government form help, GitHub commit message, PowerPoint picture insert
Benchmarks (bonus)	Latency and memory stats from Qualcomm AI Hub for chosen models
Code Repo (bonus)	GitHub link with frontend/wrapper code and qai-hub compile script

12. One-Line Pitch

ScreenSense Guardian — The on-device AI copilot that helps 40+ and elderly users learn any app, navigate any website, avoid scams, and use tools like Word, PowerPoint, GitHub, and government portals with confidence. Privately, instantly, and offline on Snapdragon-powered HP PCs.

“No more YouTube tutorials. No more asking your kids. Just ask your screen.”

13. Why This Wins

Reason	Why judges will love it
Real human problem	Not another chatbot. Solves scams, government forms, digital confidence, and learning everyday apps.
Emotional hook	Elderly scam protection + dignity of learning without embarrassment.
Universal proof	Works on government sites, banking, GitHub, VS Code, Word, PowerPoint, any screen.
Smart product design	“Show, don’t do” keeps user in control. Safe, empowering, buildable.
Perfect NPU fit	Privacy, latency, offline, battery — all require on-device AI.
Demo-friendly	Fake pop-up warning, form help, commit message suggestion, PowerPoint picture insert — all visual.

Prepared for the Qualcomm AI Lab Challenge — Snapdragon-powered HP PCs.